

YOGESHWAR ARALA

AI/ML ENGINEER | PYTHON DEVELOPER

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PROFESSIONAL SUMMARY

Recent Computer Science graduate (B.Tech, 2025) specializing in AI/ML and Python development, with hands-on experience across two machine learning internships and three applied projects spanning classical ML, deep learning, and LLM-based Retrieval-Augmented Generation (RAG) systems. Proficient in Python, Scikit-learn, TensorFlow, LangChain, and FAISS, with a consistent record of taking models from data preprocessing through evaluation and stakeholder-ready reporting.

EDUCATION

B.Tech in Computer Science Engineering

Nov 2021 – Aug 2025

Sreyas Institute of Engineering and Technology, Hyderabad | CGPA: 7.0 / 10

TECHNICAL SKILLS

Languages: Python, C, C++, Java

ML / DL: Scikit-learn, TensorFlow, PyTorch, ANN, CNN, XGBoost

Generative AI: LLMs, LangChain, RAG, Hugging Face Transformers, FAISS, ChromaDB, Sentence Transformers

Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, NLP, Feature Engineering

Tools & Practices: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Agile Collaboration, Cross-Validation, Model Evaluation

PROJECTS

DocuMind — AI-Powered Multi-Document Question Answering System (RAG)

- Engineered an end-to-end multi-document **RAG system using LangChain, FAISS, BM25, Reciprocal Rank Fusion (RRF), Gemini Embeddings, and Google Gemini**, enabling semantic + keyword retrieval across multiple PDFs.
- Implemented hybrid retrieval** by combining dense vector search with BM25 lexical search and RRF-based ranking, improving retrieval of both semantically similar queries and exact technical keywords.
- Designed grounded context augmentation and source attribution**, returning document- and page-level references while reducing unsupported or hallucinated responses through prompt-based grounding.
- Developed a production-ready Flask application with REST APIs**, interactive document viewer, retrieval metrics, and **Gunicorn cloud deployment**, enabling real-time multi-document question answering

Automated Lung Cancer Segmentation Using Deep Learning and U-Net

- Developed a lung tumor segmentation/classification pipeline on CT scans using a **Modified U-Net with residual blocks and attention gates**.
- Addressed class imbalance with a combined Dice + Cross-Entropy loss; trained/evaluated on the public LIDC-IDRI dataset with augmentation for generalization.
- Built CNN/ResNet/DenseNet classifiers for normal/benign/malignant categorization and **deployed a Flask web app** delivering real-time predictions with confidence scores

INTERNSHIP EXPERIENCE

Machine Learning Intern — Talent Trek E-Learning

Jun 2025 – Sep 2025 | Remote

- Built **end-to-end ML pipelines** in Python (Scikit-learn, TensorFlow, Pandas) covering data preprocessing, feature engineering, and hyperparameter tuning for real-world datasets.
 - Automated recurring preprocessing and tuning steps, **cutting manual iteration time** across the model development cycle.
- Delivered model evaluation reports aligned to team documentation and coding standards, supporting handoff to reviewers

Machine Learning Intern — Skill Forge

Oct 2024 – Dec 2024 | Remote

- Designed preprocessing pipelines (missing-value handling, encoding, scaling) and implemented **classification, clustering, and regression models** for predictive-analytics use cases.
- Benchmarked multiple algorithms per use case and selected the best-performing model using cross-validation.

CERTIFICATIONS

- Machine Learning Internship — Talent Trek E-Learning
- Machine Learning Internship — Skill Forge
- Python Developer Internship — DHAPPS